

assess the synthesized information, considering both quantitative data and qualitative insights, to determine optimal trading actions.

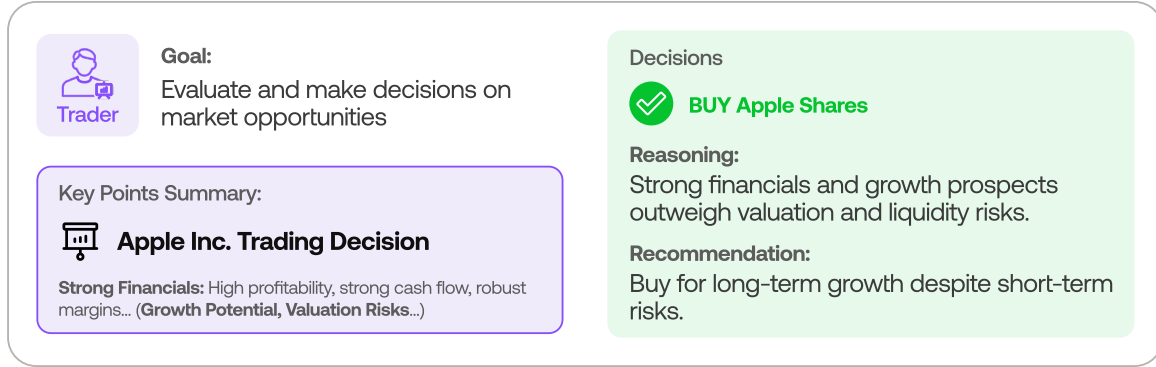


Fig. 4: TradingAgents's Trader Decision-Making Process

The tasks of TradingAgents Trader include:

- Evaluating recommendations and insights from analysts and researchers.
- Deciding on the timing and size of trades to maximize trading returns.
- Placing buy or sell orders in the market.
- Adjusting portfolio allocations in response to market changes and new information.

Trader Agents must balance potential returns against associated risks, making timely decisions in a dynamic market environment. Their actions directly impact the firm's performance, necessitating a high level of precision and strategic thinking.

3.4. Risk Management Team

The Risk Management Team (Figure 5) monitors and controls the firm's exposure to various market risks. These agents continuously evaluate the portfolio's risk profile, ensuring that trading activities remain within predefined risk parameters and comply with regulatory requirements.

The responsibilities of Risk Management Team include:

- Assessing factors such as market volatility, liquidity, and counterparty risks.
- Implementing risk mitigation strategies, such as setting stop-loss orders or diversifying holdings.
- Providing feedback to Trader Agents on risk exposures and suggesting adjustments to trading strategies.
- Ensuring that the overall portfolio aligns with the firm's risk tolerance and investment objectives.

By offering oversight and guidance, the Risk Management Team helps maintain the firm's financial stability and protect against adverse market events. They play a crucial role in safeguarding assets and ensuring sustainable long-term performance.

All agents in TradingAgents follow the ReAct prompting framework (Yao et al., 2023), which synergizes reasoning and acting. The environment state is shared and monitored by the agents, enabling them to take context-appropriate actions such as conducting research, executing trades, engaging in debates, or managing risks. This design ensures a collaborative, dynamic decision-making process reflective of real-world trading systems.

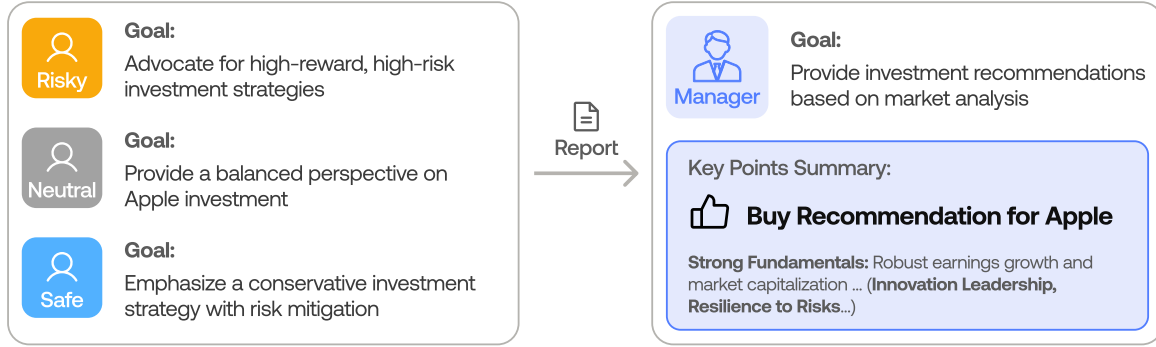


Fig. 5: TradingAgents Risk Management Team and Fund Manager Approval Workflow

4. TradingAgents: Agent Workflow

4.1. Communication Protocol

Most existing LLM-based agent frameworks use natural language as the primary communication interface, typically through structured message histories or collections of agent-generated messages (Fatouros et al., 2024b; Li et al., 2023a; Yang et al., 2023c, 2024). However, relying solely on natural language often proves insufficient for solving complex, long-term tasks that require extensive planning horizons. In such cases, pure natural language communication can resemble a game of telephone—over multiple iterations, initial information may be forgotten or distorted due to context length limitations and an overload of text that obscures critical earlier details (Hong et al., 2024). To address this limitation, we draw inspiration from frameworks like MetaGPT, which adopt a structured approach to communication. Our model introduces a structured communication protocol to govern agent interactions. By clearly defining each agent’s state, we ensure that each role only extracts or queries the necessary information, processes it, and returns a completed report. This streamlined approach reduces unnecessary steps, lowers the risk of message corruption, and keeps interactions focused and efficient, even in complex, long-horizon tasks.

4.2. Types of Agent Interactions

In contrast to previous multi-agent trading frameworks, which rely heavily on natural language dialogue, TradingAgents agents communicate primarily through structured documents and diagrams. These documents encapsulate the agents’ insights in concise, well-organized reports that preserve essential content while avoiding irrelevant information. By utilizing structured reports, agents can query necessary details directly from the global state, eliminating the need for lengthy conversations that risk diluting information, extending the message state indefinitely, and causing data loss. The types of documents and the information they contain are detailed below:

I. Analyst Team: Fundamental, sentiment, news, and technical analysts compile their research and findings into concise analysis reports specific to their areas of expertise. These reports include key metrics, insights, and recommendations based on their specialized analyses.

II. Traders: Traders review and analyze the reports from the analysts, carefully deliberating to produce clear decision signals. They accompany these decisions with detailed reports explaining their rationale and supporting evidence, which are later utilized by the risk management team.

Agents engage in natural language dialogue exclusively during agent-to-agent conversations

and debates. These concise, focused discussions have been shown to promote deeper reasoning and integrate diverse perspectives, enabling more balanced decisions in complex, long-horizon scenarios—a method particularly relevant to the intricate environment of trading (Du et al., 2023). This approach seamlessly integrates with our structured framework, as the conversation state is recorded as a structured entry within the overall agent state. The types of communication in these scenarios are detailed below:

III. Researcher Team: Each researcher agent queries the global agent state for analyst reports and carefully forms their opinion. Two researchers represent opposing perspectives: one bullish and one bearish. They engage in natural language dialogue for n rounds, as determined by the debate facilitator agent. At the conclusion, the facilitator reviews the debate history, selects the prevailing perspective, and records it as a structured entry in the communication protocol.

IV. Risk Management Team: The risk management team, similar to the researcher team, queries the trader’s decision and accompanying report. They then deliberate from three perspectives—risk-seeking, neutral, and risk-conservative—to adjust the trading plan within risk constraints. They engage in n rounds of natural language discussion, guided by a facilitator agent.

V. Fund manager: The fund manager reviews the discussion from the risk management team, determines the appropriate risk adjustments, and updates the trader’s decision and report states within the communication protocol.

4.3. Backbone LLMs

To meet the diverse complexity and speed demands of tasks in our framework, we strategically select Large Language Models (LLMs) based on their strengths. Quick-thinking models, such as gpt-4o-mini and gpt-4o, efficiently handle fast, low-depth tasks like summarization, data retrieval, and converting tabular data to text (OpenAI et al., 2024). In contrast, deep-thinking models like o1-preview excel in reasoning-intensive tasks such as decision-making, evidence-based report writing, and data analysis. These models leverage their architectures for multi-round reasoning, producing logically sound, in-depth insights (OpenAI, 2024; Wang et al., 2024a; Zhong et al., 2024). Additionally, we prioritize models with proven reliability and scalability to ensure optimal performance across various market conditions. We also employ auxiliary expert models for specialized tasks like sentiment analysis.

Specifically, all analyst nodes rely on deep-thinking models to ensure robust analysis, while quick-thinking models handle data retrieval from APIs and tools for efficiency. Researchers and traders use deep-thinking models to generate valuable insights and support well-informed decisions. By aligning the choice of LLMs with the specific requirements of each task, our framework achieves a balance between efficiency and depth of reasoning, which is crucial for effective trading strategies.

This implementation strategy ensures that TradingAgents can be deployed without requiring a GPU, relying only on API credits. It also introduces seamless exchangeability of backbone models, enabling researchers to effortlessly replace the model with any locally hosted or API-accessible alternatives in the future. This adaptability supports the integration of improved reasoning models or finance-tuned models customized for specific tasks. As a result, TradingAgents is highly scalable and future-proof, offering flexibility to accommodate any backbone model for any of its agents.